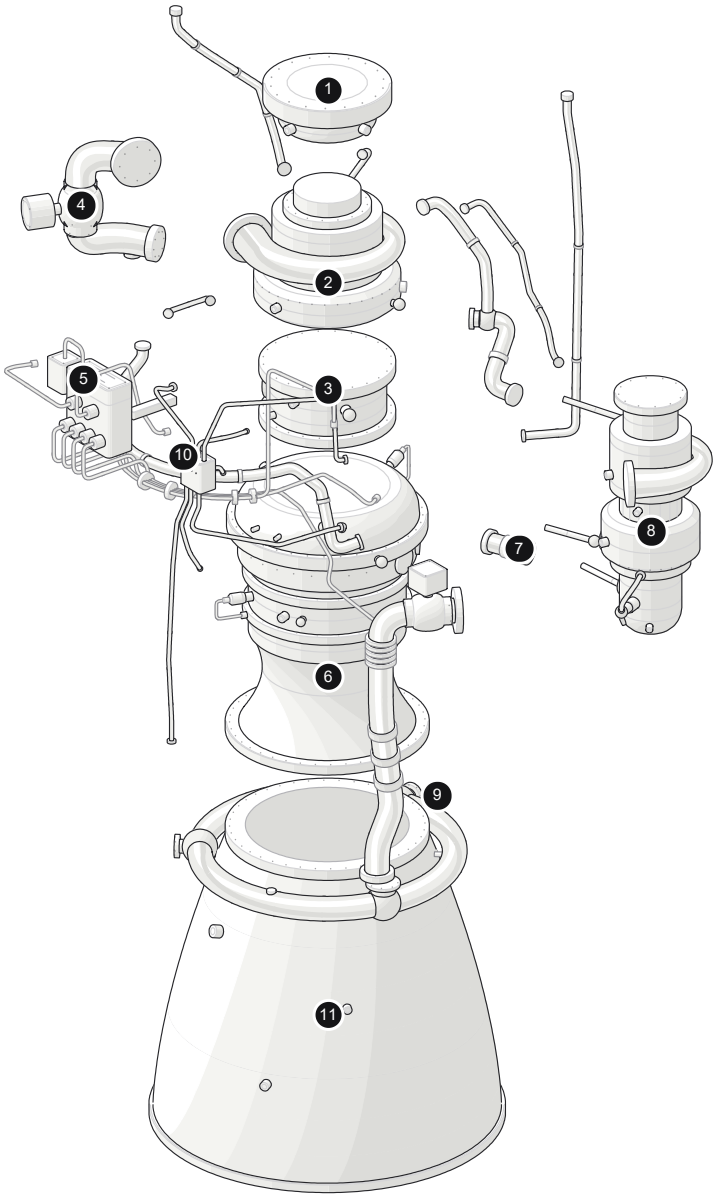
Raptor 2: how a rocket engine works

AN-001



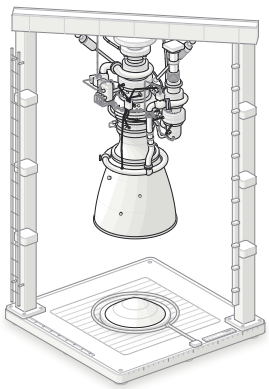
PARTS

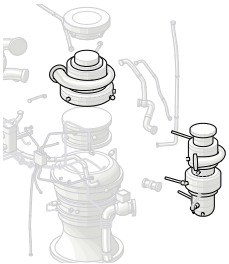
01 Gimbal and thrust puck
02 Oxygen turbopump
03 Oxygen-rich preburner
04 Oxygen duct and main valve
05 Engine controller and harness
06 Injector and main chamber
07 Hot-gas manifold
08 Methane turbopump and fuel-rich preburner
09 Downcomer and main fuel valve
10 Plumbing
11 Nozzle

SPECIFICATION

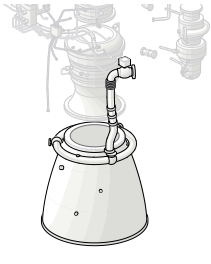
Cycle	Full-flow staged combustion
Propellants	LOX / methane
Mixture	3.6 : 1
Chamber	300 bar
Thrust	230 tf, sea level
Exit	1.3 m
Dry mass	1,630 kg

ASSEMBLED

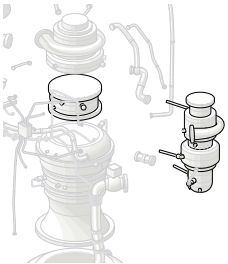




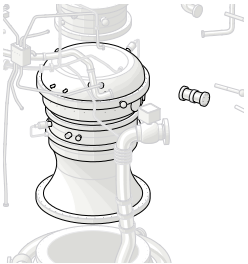
01
Pumps raise the pressure



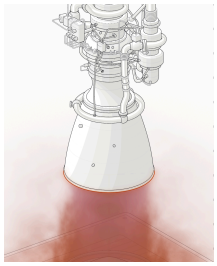
02
Methane cools the bell



03
Preburners spin the pumps



04
Hot gases meet



05
Thrust